Optimizing Revenue Leakage & Profitability

Hospitality Sector Analysis & Strategic Initiatives

Executive Summary

- Analysis of Atliq Group of Hotels across Delhi, Mumbai, Hyderabad, and Bangalore.
- Services cover 7 property types and 4 room types.
- Time period: May to July 2022 (134,590 room bookings analyzed).
- 24.83% of bookings were cancelled, driving substantial revenue leakage.
- Shifted focus from absolute 'Loss' to a standardized 'Profitability Index' (Revenue Realized / Revenue Generated).
- Average stay duration is 2 days; average lead time is 4 days.

Objectives

- Identify the root cause of substandard operational performance and revenue leakage.
- Fix data leakage in legacy data pipelines.
- Establish the Profitability Index as the primary KPI.
- Propose new, data-driven initiatives to mitigate leakage and improve revenue realization.
- Develop a real-time dashboard (Streamlit) for monitoring ongoing performance.

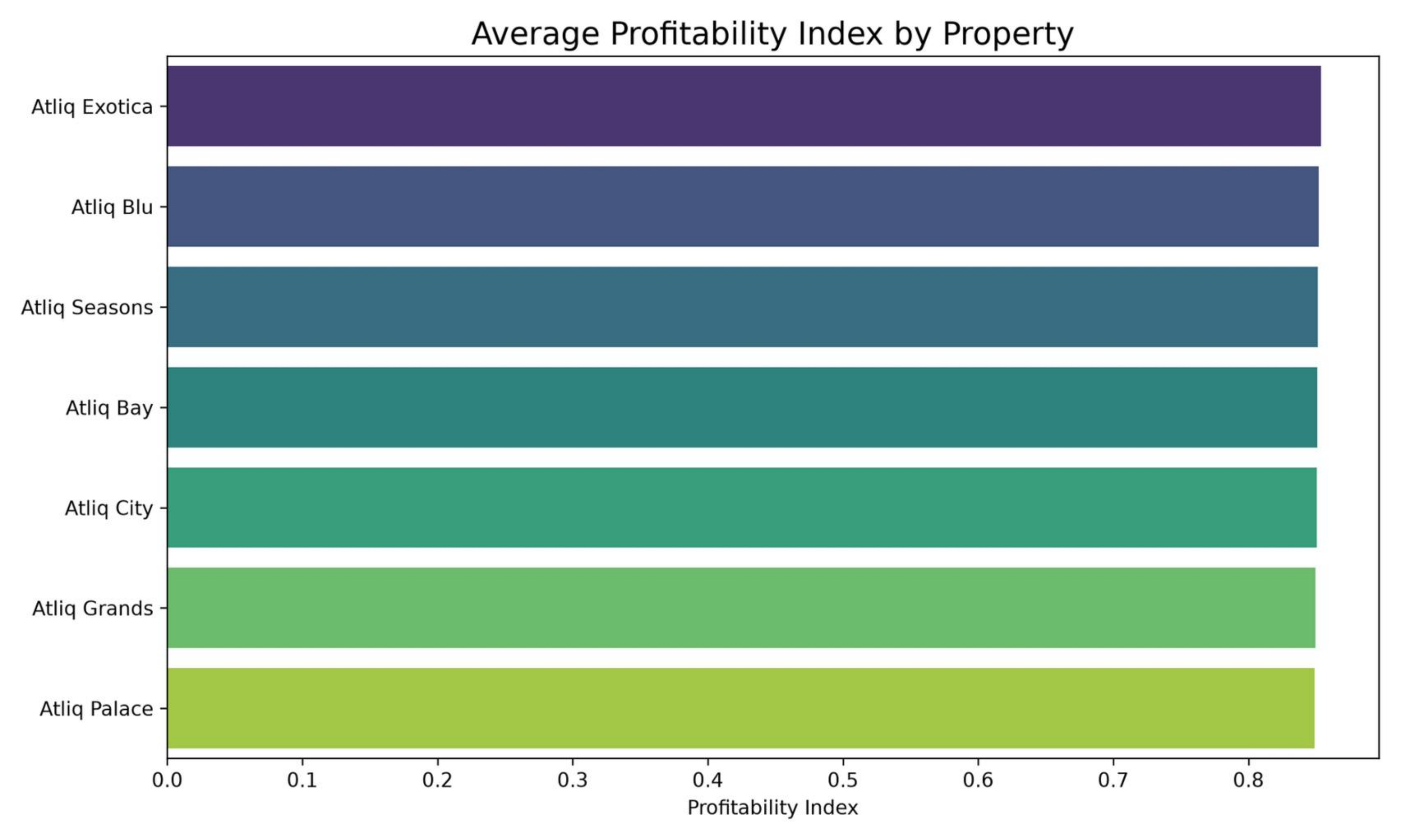
Data Sources & Preprocessing

- Consolidated multiple fact and dimension tables (bookings, aggregated bookings, hotels, rooms, dates).
- Performance Optimization: Migrated output to .parquet to improve dashboard I/O speed.
- Data Leakage Fix: Replaced predictive imputation that used future target variables (revenue) with safe, grouped median imputation for missing ratings.
- Feature Engineering: Created Profitability Index, Advance Booking window, and Length of Stay.

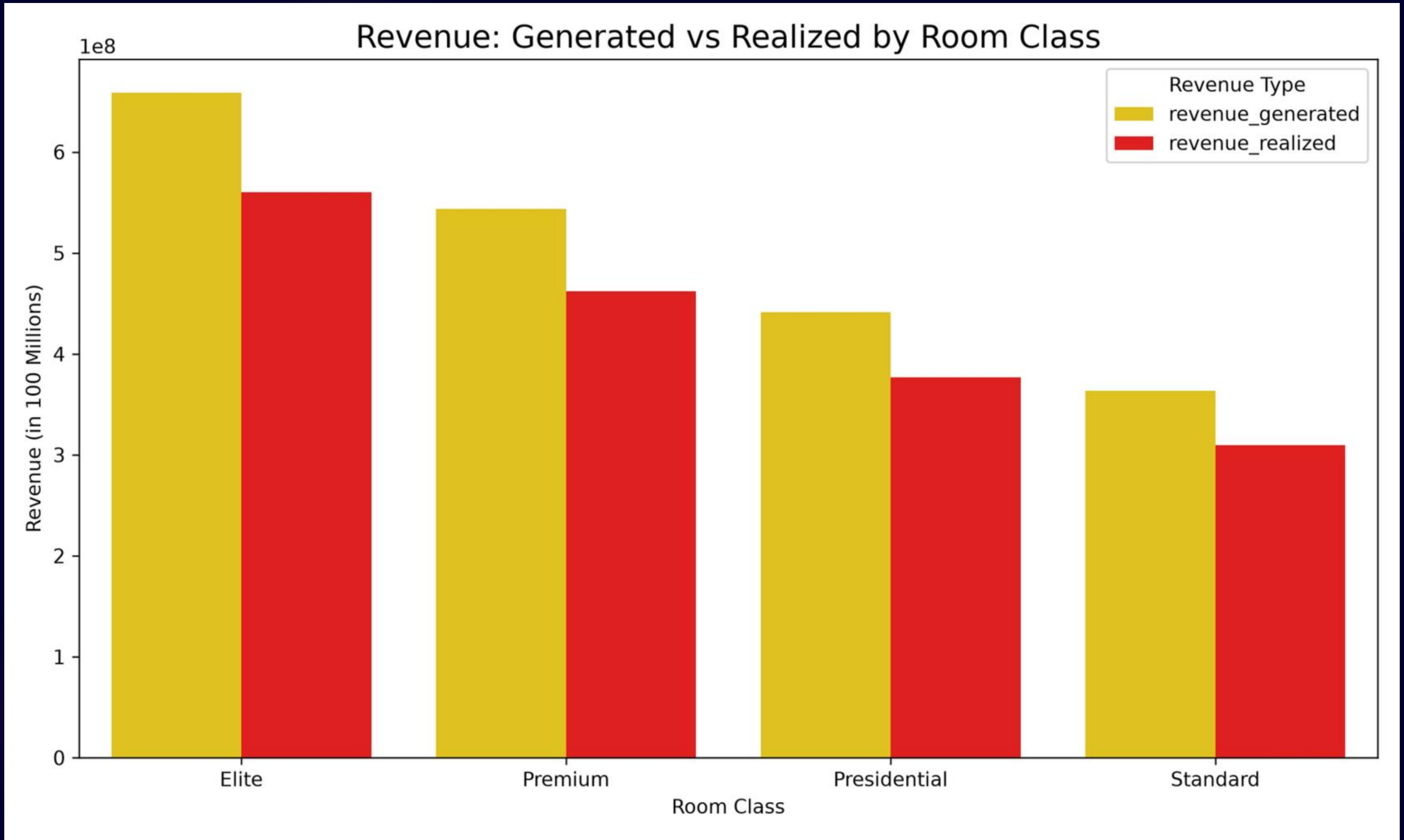
Key Performance Indicators (KPIs)

- Total Revenue Generated: Rs. 2,007,546,215
- Total Revenue Realized: Rs. 1,708,771,229
- Average Profitability Index: 85.10%
- Overall Occupancy Rate: 61.34%
- Average Guest Rating: 3.81

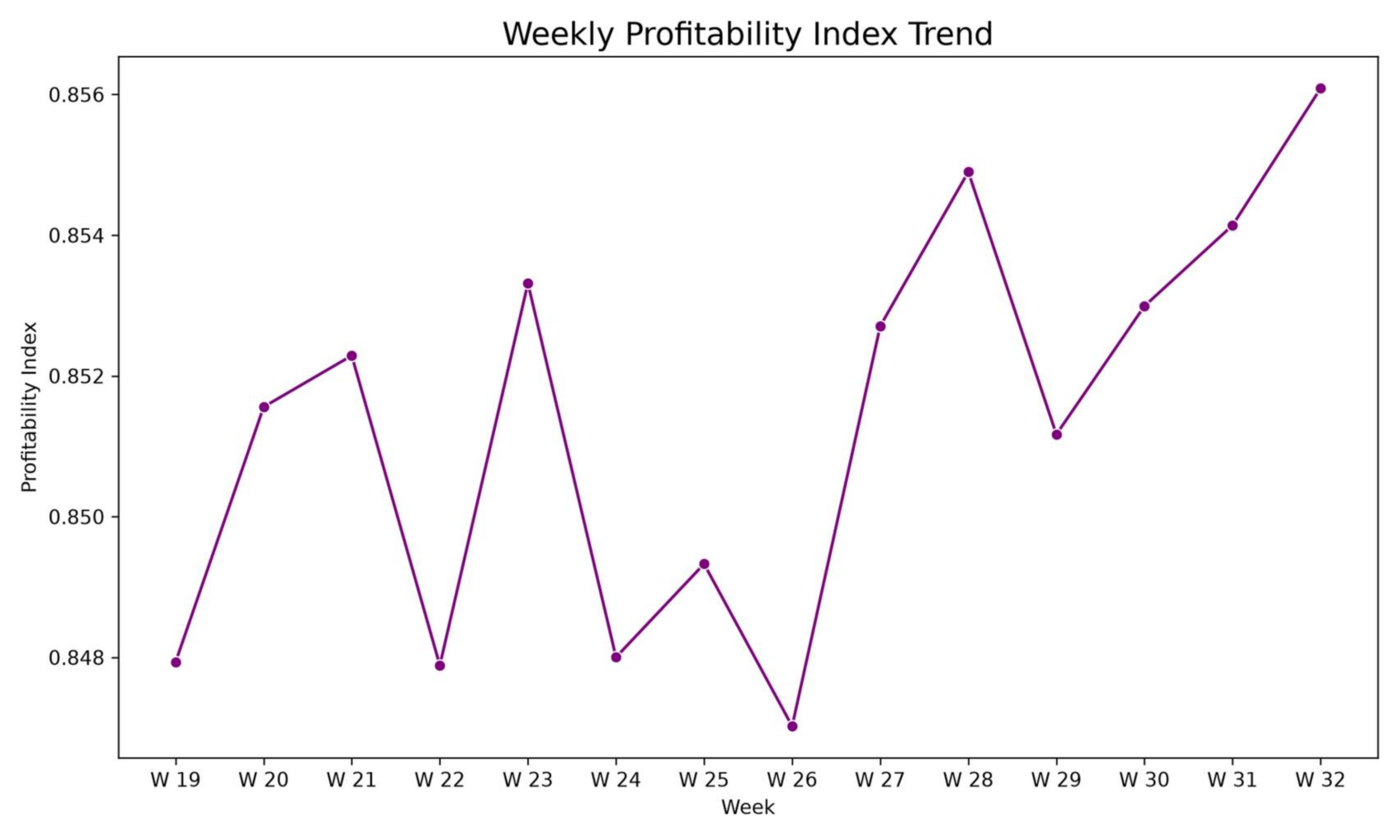
Profitability Index by Property



Revenue Realized vs Generated by Room Class



Weekly Profitability Trend



Root Cause & Operational Analysis

- Major revenue leakage observed in specific room types (e.g., Elite) due to cancellations.
- Weekdays experience higher leakage compared to weekends.
- Third-party booking platforms have significantly higher cancellation rates than direct booking (LogTrip).
- Pricing is not dynamic, leaving capacity underutilized during off-peak periods.

Proposed Changes & New Initiatives

- Tighten Cancellation Rules on Booking Platforms (Estimated ₽25k/mo incremental revenue).
- Implement Dynamic Pricing Engine to maximize realization during high demand (Estimated ₽45k/mo).
- Launch Direct Booking Mobile App & Loyalty Program to bypass third-party commissions.
- Increase minimum stay on weekends to protect capacity.
- Offer Last-Minute Discounts for 1-Night stays to capture perishable inventory.

Conclusion & Key Takeaways

- Data Leakage Resolved: Reporting and modeling are now structurally sound.
- Revenue Leakage is concentrated: Targeted policy changes can yield immediate ROI.
- High-Impact Initiatives: Dynamic pricing and direct booking channels offer the strongest long-term payback.
- Dashboard Deployed: A new Streamlit dashboard provides stakeholders with real-time tracking of the Profitability Index.